Transformers Architecture

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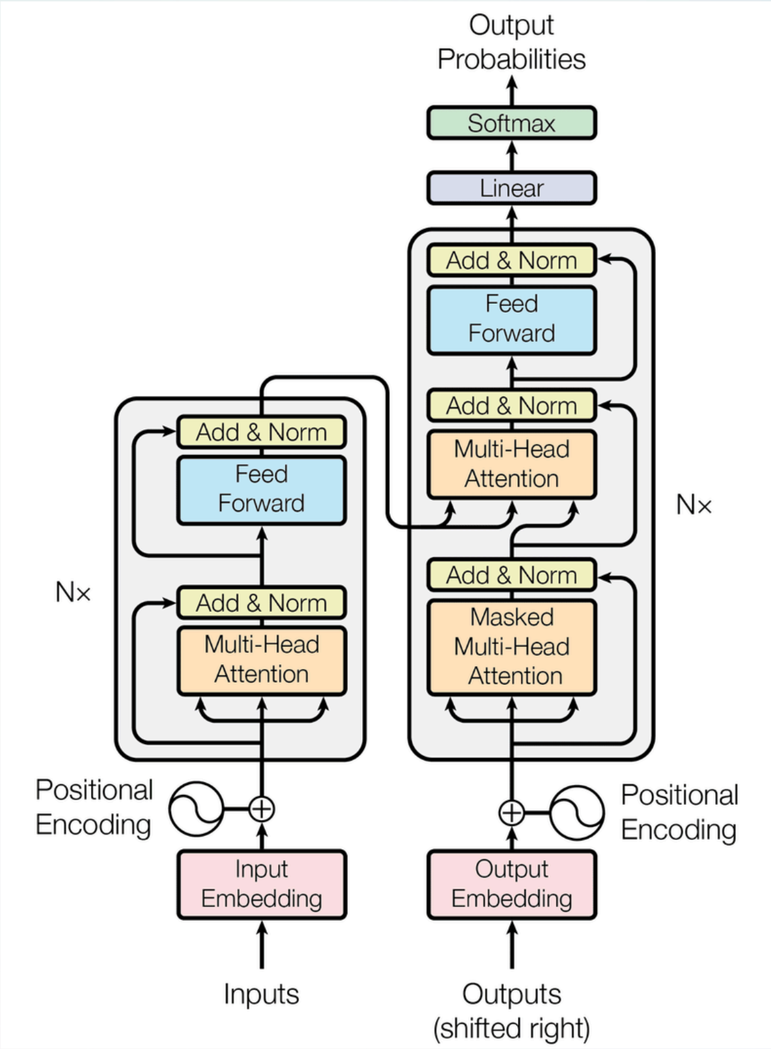
Encoder Architecture

Decoder Architecture

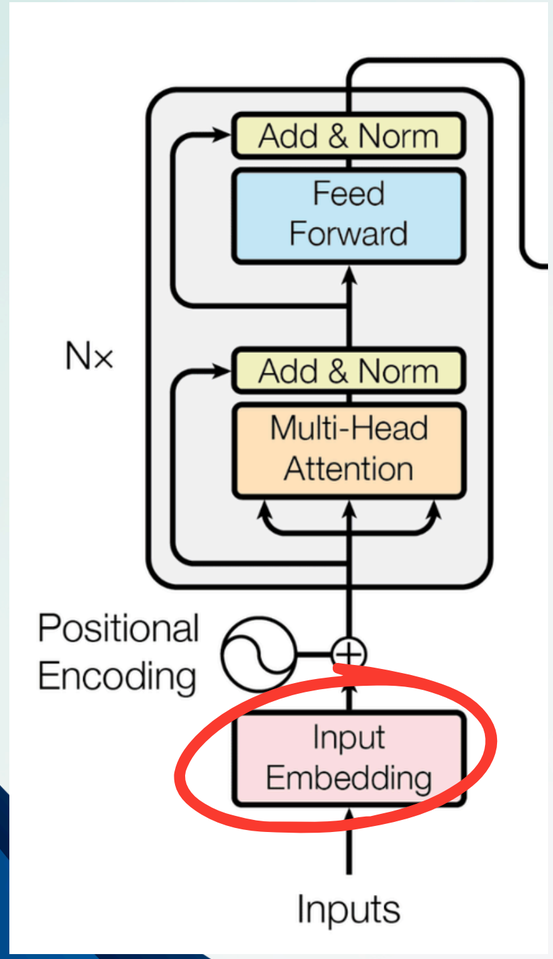
ChemBERTa Architecture

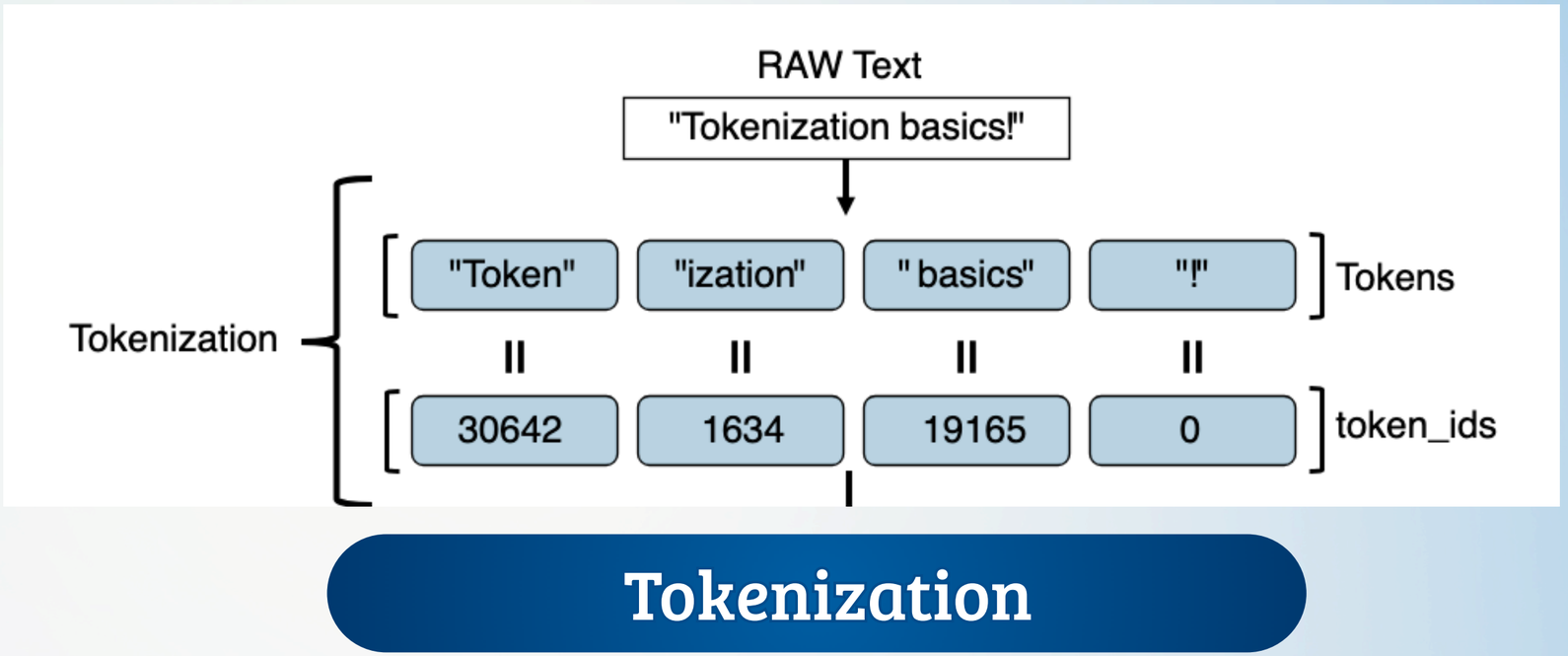
MoL-Gen Architecture

Transformer Architecture



Input Embedding





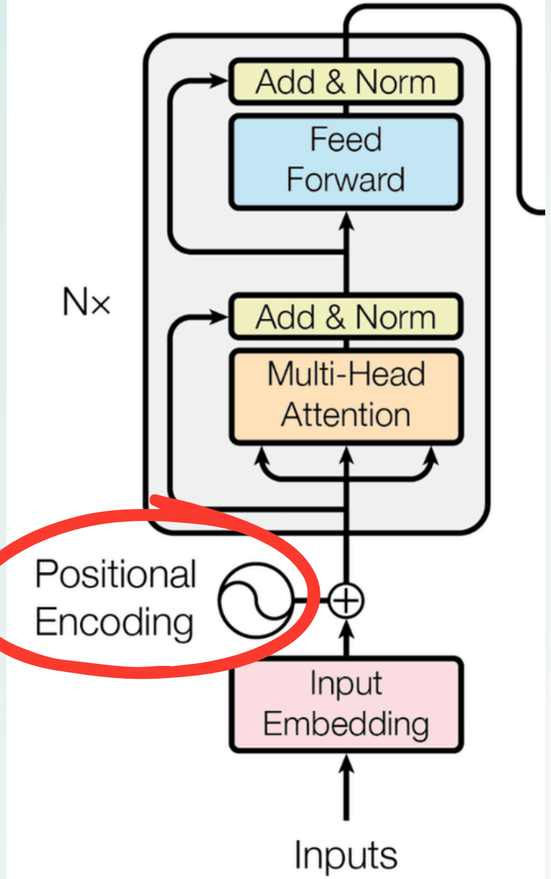
Tokenization

Input Embedding

hello	12	45	43	26	78	532	...
there	43	25	778	43	53	78	...
texas	34	56	23	12	56	74	...
world	342	54	23	5	7	423	...
...	...						

Embedding

Positional Embedding



$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right)$$

Positional Embedding Formula

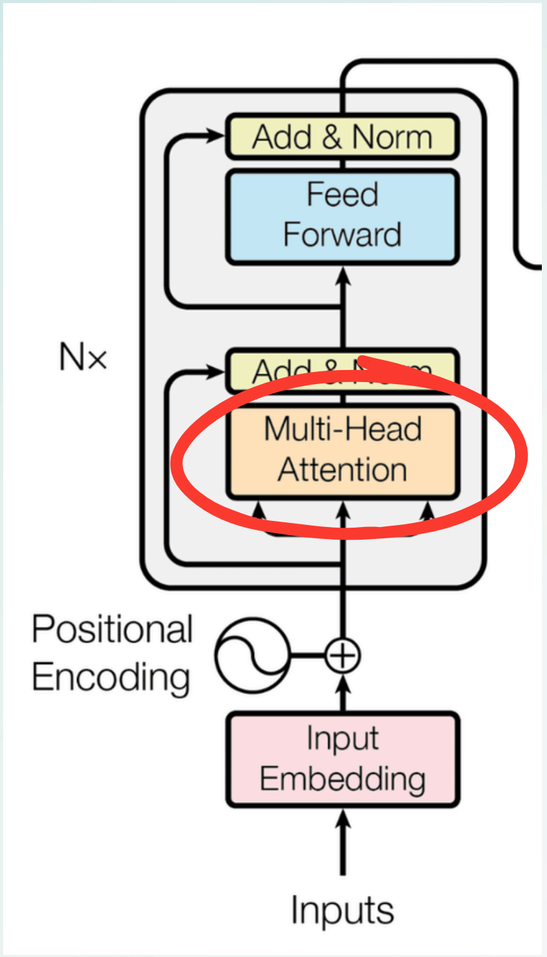
Positional Embedding

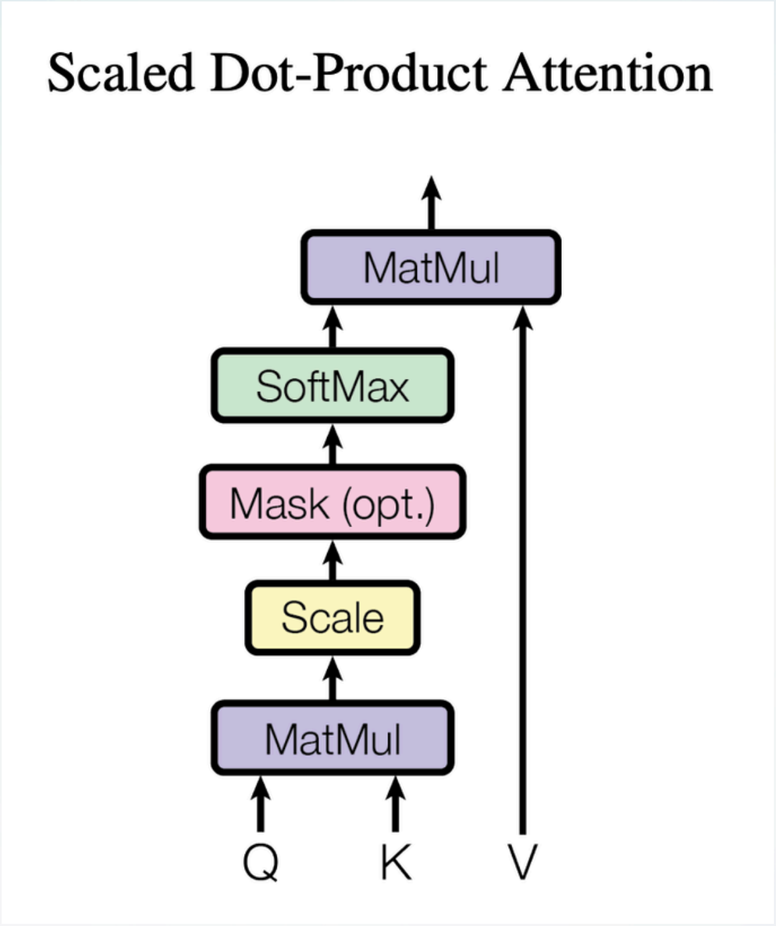
In this work, we use sine and cosine functions of different frequencies:

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

where pos is the position and i is the dimension. That is, each dimension of the positional encoding corresponds to a sinusoid. The wavelengths form a geometric progression from 2π to $10000 \cdot 2\pi$. We chose this function because we hypothesized it would allow the model to easily learn to attend by relative positions, since for any fixed offset k, PE_{pos+k} can be represented as a linear function of PE_{pos}.

Self-Attention





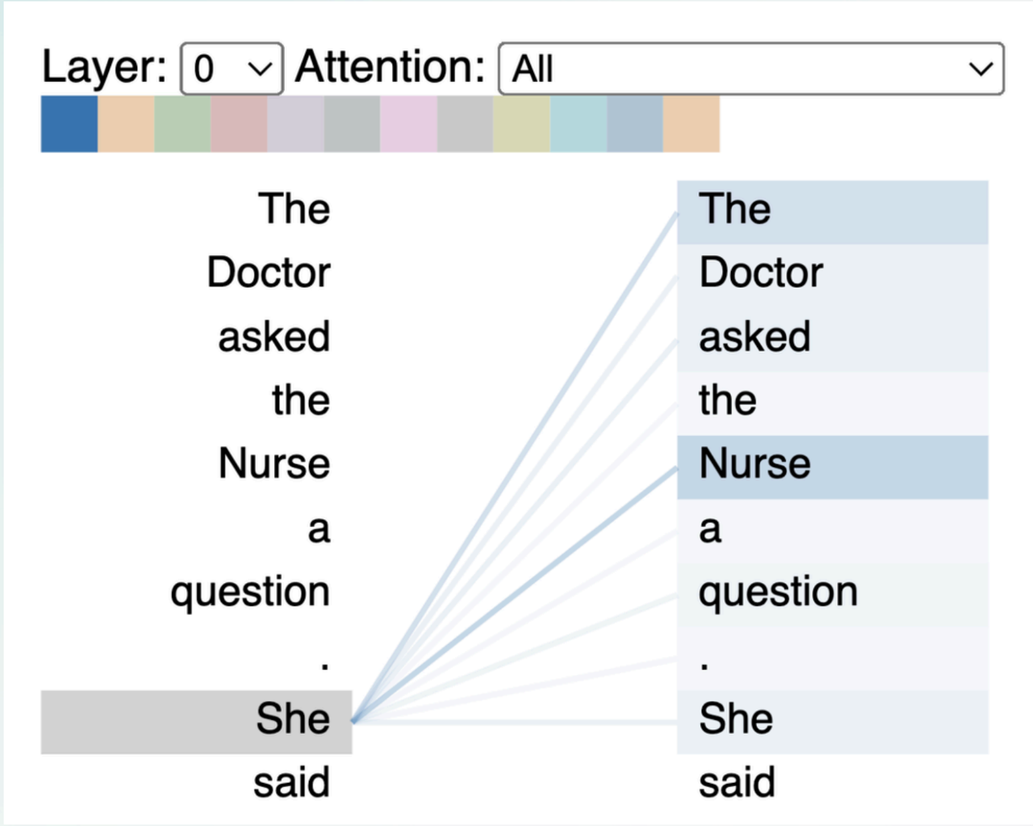
$$\mathbf{q}_n = \boldsymbol{\beta}_q + \boldsymbol{\Omega}_q \mathbf{x}_n$$

$$\mathbf{k}_m = \boldsymbol{\beta}_k + \boldsymbol{\Omega}_k \mathbf{x}_m$$

$$\mathbf{v}_m = \boldsymbol{\beta}_v + \boldsymbol{\Omega}_v \mathbf{x}_m$$

Query: What am I looking for
Key: Response to the Query
Value: Actual Information

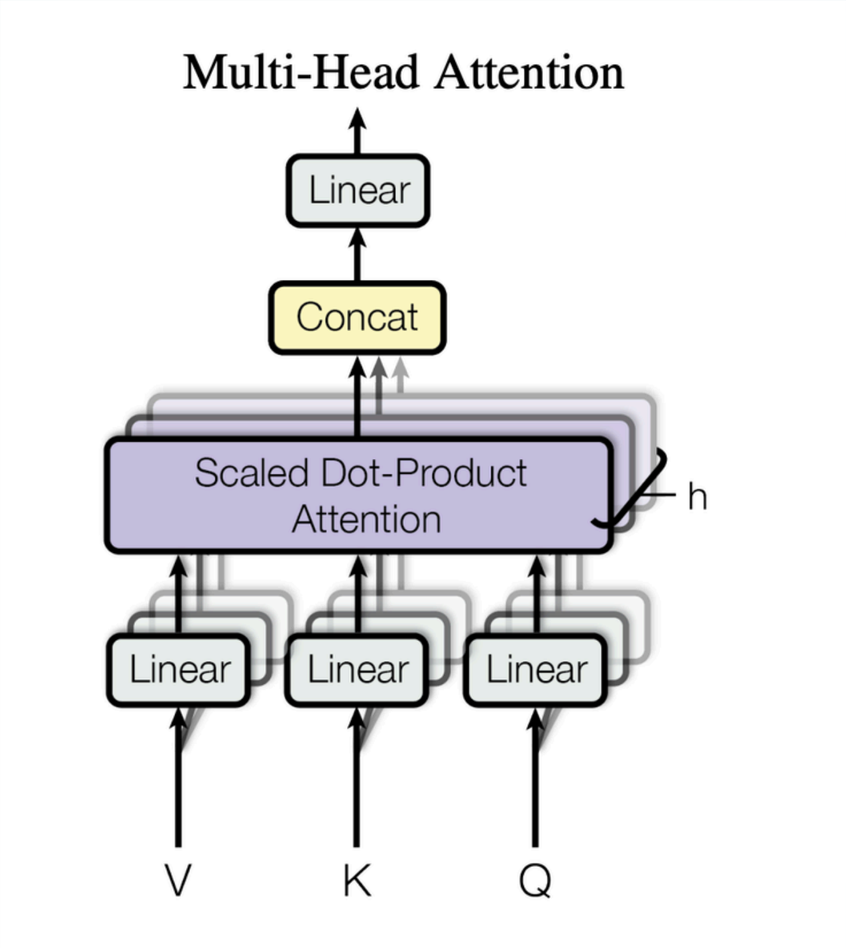
Self-Attention



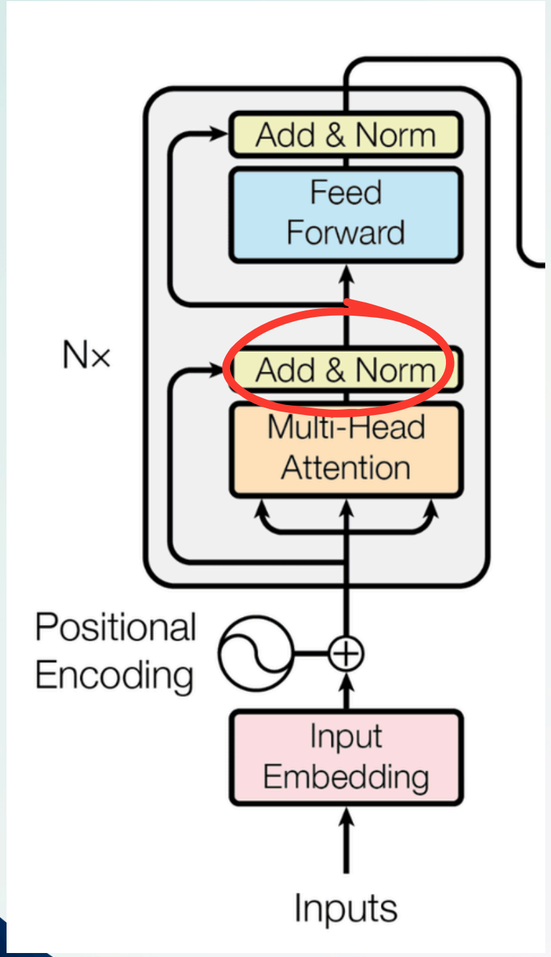
$$a[\mathbf{x}_m, \mathbf{x}_n] = \text{softmax}_m\left[\frac{\mathbf{k}_m^T \mathbf{q}_n}{\sqrt{d_k}}\right]$$

$$\text{sa}_n[\mathbf{x}_1, \ldots, \mathbf{x}_N] = \sum_{m=1}^{N} a[\mathbf{x}_m, \mathbf{x}_n]\, \mathbf{v}_m$$

Multi-Head Self-Attention

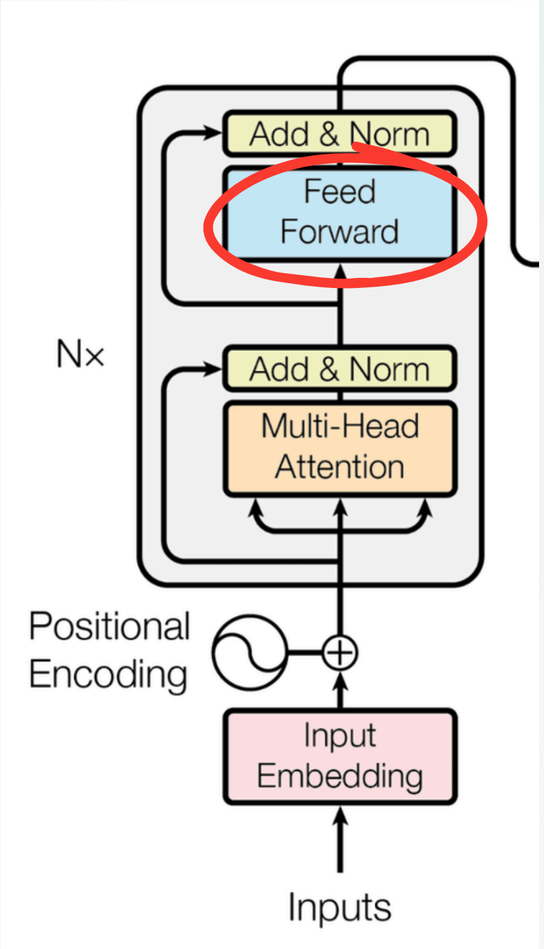


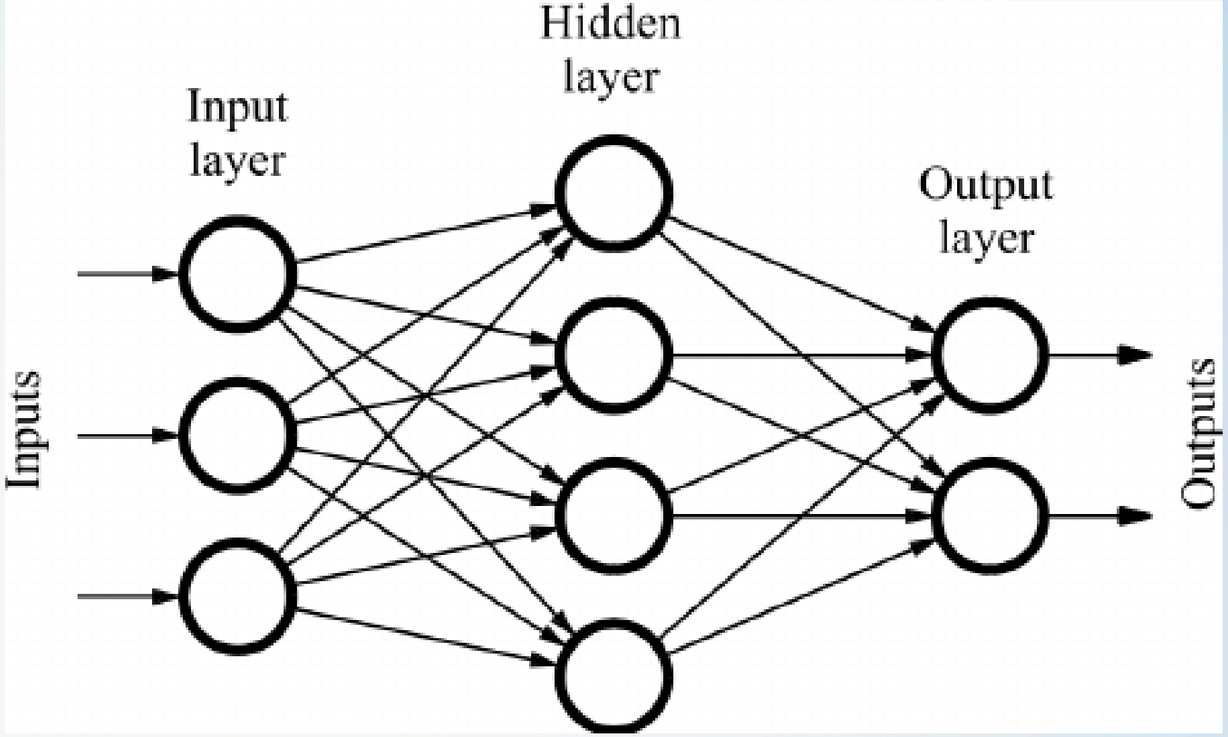
Layer Normalization



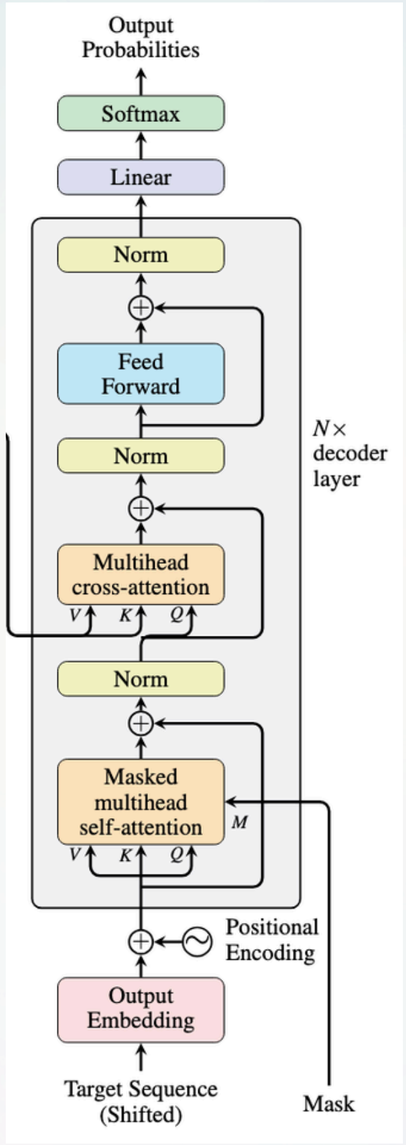
$$\mathbf{LN}(\mathbf{x}) = \boldsymbol{\gamma} \odot \frac{\mathbf{x} - \mu}{\sqrt{\sigma^2 + \epsilon}} + \boldsymbol{\beta}$$

Feed Forward

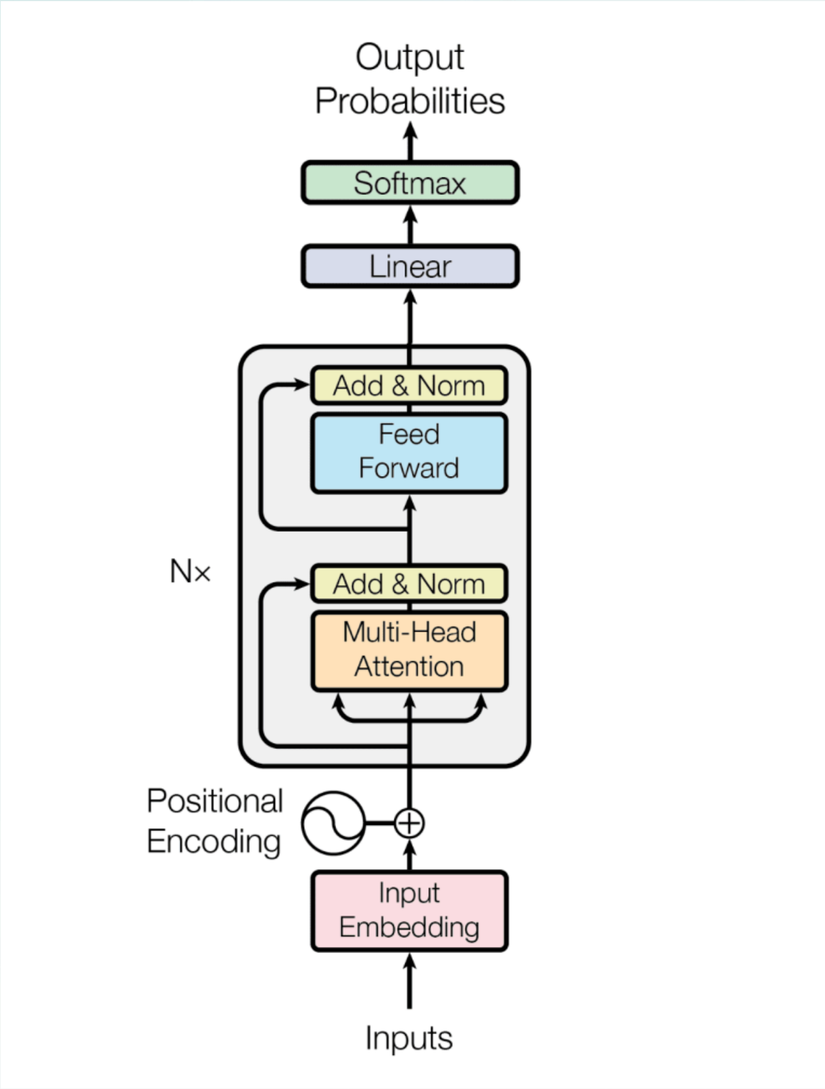




Decoder Architecture

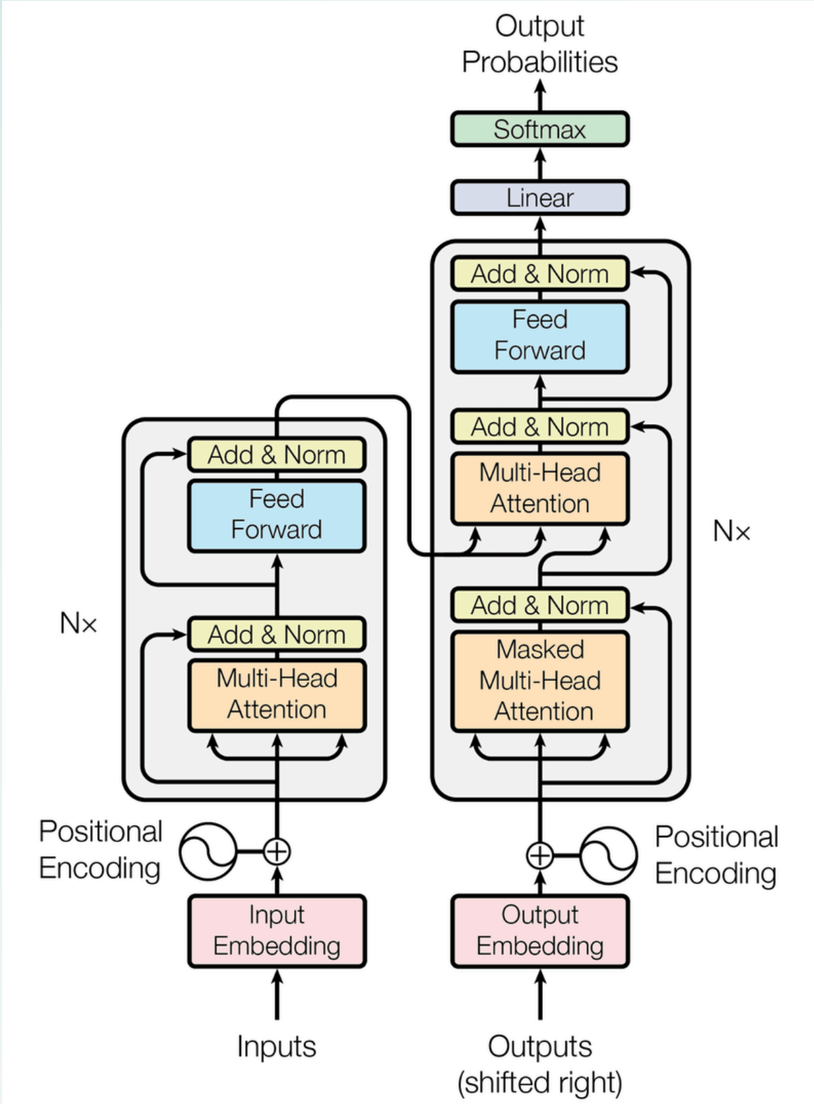


ChemBERTa Architecture



BERT (Bidirectional Encoder Representations from Transformers)

MoL-Gen Architecture



BART (Bidirectional and Auto-Regressive Transformers)

THANK YOU